DRILL 07

HOSE & STREAMS

NFPA 1001 Level 1 (JPR 4.3.10) / WorkSafeNB Reg 91-191

HOSE HANDLING, BODY MECHANICS & ADVANCEMENT

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Nozzleman & Backup Firefighter

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must master modern hose handling mechanics (the **Hip Grip** or **Arm-Clamp** technique) to absorb 60–80 lbs of nozzle reaction force without fatigue, coordinate advancement with a backup firefighter, negotiate 90-degree corners and stairs, and maintain strict door control. Proper technique prevents fatigue and prevents nozzlemen from being thrown off balance.

📦 TOOLS & APPARATUS SETUP

- 1 charged 1.75" (45mm) attack line (100–150 ft) flowing at normal fireground operating pressure (~100–125 PSI discharge pressure, delivering 150–160 GPM / 600 LPM).
- Simulated corner/hallway (between apparatus bays, around building corners, or training door prop).
- Webbing strap (for door control).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Back Strain & Ergonomics:** Watch for nozzlemen leaning backward or fighting nozzle reaction with bent arms. The nozzle reaction must be driven straight down into the hips and legs.

⚠️ **Pinch Points & Knuckles:** Ensure hands are not dragged along brick/door frames when navigating tight corners under pressure.

⚠️ **Water Runoff:** Direct runoff toward exterior storm drains or open gravel; monitor for slick surfaces in freezing weather.

🛑 **THE STOP RULE: Shout "STOP!" if the nozzle firefighter loses grip or control of the bail, or if a member falls.**

PHASE 1: MODERN NOZZLE MECHANICS (THE HIP GRIP) (15 MINUTES) — SLOW DRY RUN

1. **The Flaw of the "Death Grip":**
   - Grasping the nozzle with two hands held out in front of the chest burns out shoulder and arm muscles in under 2 minutes.
2. **The Modern Hip Grip Technique:**
   - Tuck the charged hose line firmly between your hip and upper arm (or lock it under your armpit against your ribcage).
   - Rest your forward hand on the hose 6–8 inches behind the nozzle coupling.
   - Rest your rear hand on top of the nozzle shutoff bail.
   - Lean forward aggressively with your front knee bent and back leg braced (athletic boxer stance).
   - The backward thrust of the water stream is absorbed directly by your core and hips into the ground.

DRILL 07: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Hose & Streams • Practical Application

PHASE 2: THE BACKUP FIREFIGHTER'S ROLE (15 MINUTES) — BUDDY COACHING

1. **Positioning:**
   - The backup firefighter positions on the **same side of the hose** as the nozzleman, 3 to 4 feet behind.
   - Lock your forward arm around the hose and rest your back hand firmly against the nozzleman's SCBA cylinder harness or shoulder.
2. **Absorbing Reaction & Pushing Forward:**
   - The backup absorbs any remaining backward reaction, allowing the nozzleman to direct the stream freely with minimal effort.
   - When moving forward, the backup firefighter provides the forward drive: *"Moving forward in 3, 2, 1... Push!"*

PHASE 3: CORNER NAVIGATION & DOOR CONTROL ADVANCEMENT (20 MINUTES) — TIMED / STRESS

1. **Navigating a 90-Degree Corner:**
   - Never let the hose pin the nozzleman against the outside wall of a corner.
   - The backup firefighter drops to one knee at the corner vertex, becoming a **"human pulley"**, holding the hose off the wall so slack feeds freely to the nozzleman.
   - Nozzleman advances 10–15 feet into the room, sweeps the ceiling and floor, and calls for more line.
2. **Door Control & Chocking:**
   - Before entering, firefighter checks the door for heat with gloved hand/TIC.
   - Chock the door open using a wooden wedge or hose strap so the door cannot pinch the charged hose and cut off water!

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Standing upright while flowing:* Standing tall raises your center of gravity; nozzle reaction will push you backwards. Stay low and lean into the stream.
- ❌ *Backup firefighter on opposite side:* Straddling the hose or standing on opposite sides creates a scissors hazard if the line whips or snags.
- ❌ *Letting the door close on the hose:* A self-closing spring hinge will pinch a charged 1.75" line against the jamb, choking water flow to a trickle. Always chock the door!

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"How big of a difference did the hip grip make in muscle fatigue?", "How well did the backup firefighter communicate forward movement?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Nozzle closed smoothly to prevent water hammer.
- ☐ Pump throttle brought to idle; discharge gate closed.
- ☐ Hose drained, rolled, and repacked.
- ☐ Door wedges returned to gear pockets.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS